What if Mass Eats the Vacuum?

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Abstract

This paper introduces a classical field theory of gravity built on the Maxwell equations and a central physical hypothesis: that mass absorbs radiation, inducing a net inward momentum flow. We assume a Lorentz-invariant stochastic zero-point radiation field present throughout space and hypothesize that mass acts as a sink for this field. We construct a consistent model of a massive particle, demonstrate energy and momentum conservation, and show that the three classic tests of general relativity: Mercury's precession, gravitational lensing, and gravitational redshift are reproduced to leading order. This approach provides a novel, Lorentz-invariant, classical alternative to general relativity, and lays a foundation for further exploration.

1 Introduction

Modern physics rests on two towering achievements: the Standard Model, underpinned by the Dirac equation, and General Relativity, based on Einstein's field equations. One governs the small, the other the large. Each is remarkably successful in its domain.

But they don't fit together. When applied simultaneously, the equations break down and yield infinities. Proposals like string theory offer elegant mathematics but few concrete predictions. Despite nearly a century of effort, no unifying framework has emerged. A fresh approach is warranted.

This paper proposes that fresh approach. Rather than quantizing Einstein's field equations or geometrizing the Dirac equation, we suggest something simpler and stranger: what if mass eats the vacuum?

1.1 Roadmap

This paper is structured to follow the logic of the scientific method: we begin with a hypothesis, outline the assumptions that support it, explore its mathematical consequences, compare its predictions to experiment, and finally examine whether it offers a deeper conceptual shift.

In **Section 2**, we introduce the hypothesis: that mass acts as a sink for radiation and in particular from a pervasive stochastic zero-point radiation field. We first present the idea intuitively, anchoring the theory in a physical picture, and then spell it out mathematically.

In **Sections 3 and 4**, we lay out the two assumptions underlying the hypothesis. By "assumptions," we mean elements already present in the scientific literature, perhaps unconventional but not novel to this work. **Section 3** introduces the stochastic zero-point radiation (Z.P.R.) field as the physical manifestation of the vacuum. **Section 4** adopts the Maxwell equations as the correct mathematical framework for describing gravity.

Sections 5 and 6 contain the technical core of the paper. We introduce a toy model of a massive particle and discuss characteriscts of the toy model. We examine whether the inflow of radiation can stabilize mass against gravitational self-energy and show that the energy needed for stability of our toy model is less than that available in the Z.P.R. We also demonstrate conservation of energy and momentum and several other key mathematical results.

In **Section 7**, we test the theory against observation. We demonstrate that it reproduces the three classic predictions of general relativity to leading order: perihelion precession, gravitational lensing, and redshift.

Section 8 explores a surprising result of the theory. And finally, **Section 9** concludes with a discussion of open questions, potential avenues for further investigation, and the broader implications of the framework proposed here.

2 The Hypothesis

In this section, we introduce the hypothesis of this paper. Before proceeding, I want to clarify the distinction between assumptions and hypothesis as used here:

- **Assumptions** are drawn from existing theoretical frameworks (even if unconventional)
- **Hypothesis** refers to the novel claim proposed in this work

Every theory begins with a hypothesis. If it yields interesting physics and remains consistent with experiment, it deserves serious consideration.

Here we first present the hypothesis intuitively, anchoring the theory in a physical picture, and then spell it out mathematically.

2.1 A Physical Picture: Radiation Sink Intuition

Before introducing equations, we present a simple physical picture to guide intuition. Consider a neutral, perfectly conducting iron sphere, heated until it glows red. When placed in the vacuum of space, it emits infrared radiation and cools down, eventually approaching absolute zero.

But at zero Kelvin, both classical and quantum theories suggest the presence of a background electromagnetic field called the stochastic zero-point radiation (Z.P.R.) field. A fuller treatment of the Z.P.R. will be offered in **Section 3**, but for now, it is enough to imagine a low-level radiation noise present even in the coldest vacuum. The iron sphere, after it has come into equilibrium with the vacuum, then absorbs radiation from this ambient field.

The hypothesis proposed here is that mass absorbs energy from the zero-point field even at absolute zero. This interaction produces a net inward flow of momentum toward the mass.

The remainder of the paper builds from this intuition and develops the consequences of this idea.

2.2 Hypothesis: The Math

The image above anchors the theory.

If mass always interacts with the zero-point field, then this interaction must have physical consequences. We propose it manifests as an inward-pointing momentum density:

Comment: Is density the right word here?

$$\boxed{\mathbf{P}(r) = -\hat{\mathbf{r}}\frac{Gm\hbar}{r^2}} \tag{1}$$

A photon traveling through this field experiences a cumulative momentum shift:

$$\Delta \mathbf{p} = \int_{\text{path}} \mathbf{P}(r)\, dl \tag{2}$$

This is the hypothesis of the paper: mass creates a momentum flux capable of acting on radiation. The implications of this hypothesis will be tested in **Section 7**.

3 Zero Point Radiation (ZPR)

We now introduce our first assumption: the **stochastic zero-point radiation field**. "Stochastic" means random, in a way that will be clearly defined below, "zero-point" refers to zero-temperature and "radiation" in this context simply means light.

Historically, the Michelson–Morley experiment aimed to detect the luminiferous aether, the medium through which light was thought to propagate. The null result helped solidify the notion of a truly empty vacuum. In contrast, the stochastic zero-point radiation field treats the vacuum as filled with fluctuating electromagnetic fields, even at absolute zero.

This theory was first developed by Boyer last century but apparently has more researcher interest as of late. Stochastic zero-point radiation explains how classical systems can remain stable even while radiating energy. It has been shown to reproduce:

- The Planck blackbody spectrum (Boyer, 1969) [**?**]
- Harmonic oscillator ground-state energy $\frac{1}{2}\hbar\omega$ [**?**]
- Casimir and van der Waals forces [**?**]
- Certain features of the hydrogen atom's ground state [**?**]
- The Unruh effect [**?**]

Taken together, these results support the view, held by Boyer and others, that the vacuum is in fact an active medium, filled with fluctuating modes that persist even at $T = 0$. Importantly, the above results are achieved *without quantization*. All results are derived using classical analysis.

We therefore, assume that the vacuum is filled with a classical, Lorentz-invariant random electromagnetic radiation field even at absolute zero. As pointed out earlier, here when we say "Assumption" we mean we posit that a seperate theory is useful to this theory, we do not try and prove its claims here, we only assume that it is correct.

3.1 The Stochastic Zero-Point Radiation Field

In the theory of Z.P.R., the vacuum is not silent. It carries a random, omnipresent electromagnetic field with spectral energy density:

$$\rho(\omega) = \frac{1}{2}\hbar\omega \tag{3}$$

The zero-point radiation field plays a central role in the framework. The core hypothesis of the theory is that mass interacts with this Z.P.R. field by absorbing momentum from it.

3.1.1 Mathematical Definition

The field is stochastic and Gaussian. The electric and magnetic fields are defined as:

$$\begin{aligned}
\mathbf{E}(\mathbf{x},t) &= \sum_{\lambda=1}^{2}\int d^3k\,\hat{\epsilon}(\mathbf{k},\lambda)\sqrt{\frac{\hbar\omega}{8\pi^2}}W_\lambda(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + \text{c.c.} \\
\mathbf{B}(\mathbf{x},t) &= \sum_{\lambda=1}^{2}\int d^3k\,\frac{\mathbf{k}\times\hat{\epsilon}(\mathbf{k},\lambda)}{|\mathbf{k}|}\sqrt{\frac{\hbar\omega}{8\pi^2}}W_\lambda(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + \text{c.c.}
\end{aligned} \tag{4}$$

where $\hat{\epsilon}(\mathbf{k},\lambda)$ are polarization vectors satisfying:

$$\hat{\epsilon}\cdot\mathbf{k} = 0, \quad \hat{\epsilon}_1\cdot\hat{\epsilon}_2 = 0, \quad |\hat{\epsilon}_\lambda| = 1 \tag{5}$$

and $W_\lambda(\mathbf{k})$ are complex Gaussian random variables with correlation properties:

$$\begin{aligned}
\langle W_\lambda(\mathbf{k})\rangle &= 0 \\
\langle W_\lambda(\mathbf{k})W^*_{\lambda'}(\mathbf{k}')\rangle &= \delta_{\lambda\lambda'}\delta^3(\mathbf{k}-\mathbf{k}') \\
\langle W_\lambda(\mathbf{k})W_{\lambda'}(\mathbf{k}')\rangle &= 0
\end{aligned} \tag{6}$$

where $\langle\cdot\rangle$ denotes the stochastic average over the ensemble of random field realizations. This background field is homogeneous, isotropic, and statistically stationary, with the full ensemble remaining invariant under Lorentz transformations.

3.1.2 Note on Averaging Conventions

Before proceeding, we clarify an important distinction between classical Jackson electrodynamics and SED:

- **Jackson (time-averaging):** Fields are assumed to have time dependence $e^{-i\omega t}$, and physical quantities like the Poynting vector are obtained by time-averaging. For fields $\mathbf{E} = \mathbf{E}_0 e^{-i\omega t}$ and $\mathbf{B} = \mathbf{B}_0 e^{-i\omega t}$, the time-averaged Poynting vector is $\langle\mathbf{S}\rangle_{\text{time}} = \frac{c}{8\pi}\text{Re}(\mathbf{E}\times\mathbf{B}^*)$, where the complex conjugate arises from the time-averaging procedure.
- **SED (stochastic averaging):** Fields contain stochastic variables $W_{l,m}(\omega)$ and we average over the ensemble of random field realizations. We write

$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ where $\mathbf{E}_+$ contains W and $\mathbf{E}_+^*$ contains W^*. The stochastic average $\langle \mathbf{E} \times \mathbf{B} \rangle$ involves terms like $\langle \mathbf{E}_+ \times \mathbf{B}_+^* \rangle$, where the correlation function $\langle W_{l,m}(\omega) W^*_{l',m'}(\omega') \rangle = \delta_{ll'} \delta_{mm'} \delta(\omega - \omega')$ determines which terms survive.

Also, Boyer talks about this a lot and points to a paper by Einstein and Hopf.

3.1.3 Derivation of Energy per Mode

We now verify that this field definition gives the correct energy spectrum. We need to evaluate $\langle \mathbf{E}^2 \rangle$ and $\langle \mathbf{B}^2 \rangle$. Let us focus on $\langle \mathbf{E}^2 \rangle$ as the calculation for the magnetic field is identical.

Write $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ where $\mathbf{E}_+$ is the positive-frequency part (containing W_λ but not W_λ^*). Then:

$$\mathbf{E}^2 = (\mathbf{E}_+ + \mathbf{E}_+^*)^2 = \mathbf{E}_+^2 + 2\mathbf{E}_+ \cdot \mathbf{E}_+^* + (\mathbf{E}_+^*)^2 \tag{7}$$

Taking the stochastic average:

$$\langle \mathbf{E}^2 \rangle = 2 \langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle \tag{8}$$

The $\mathbf{E}_+^2$ and $(\mathbf{E}_+^*)^2$ terms vanish because $\langle WW \rangle = 0$, by equation (6). Evaluating the remaining term:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \sum_{\lambda,\lambda'} \int d^3k \, d^3k' \, \hat{\epsilon}_\lambda \cdot \hat{\epsilon}^*_{\lambda'} \frac{\hbar\sqrt{\omega\omega'}}{8\pi^2} \langle W_\lambda(\mathbf{k}) W^*_{\lambda'}(\mathbf{k}') \rangle e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)} e^{-i(\mathbf{k}'\cdot\mathbf{x} - \omega' t)} \tag{9}$$

Using the correlation function $\langle W_\lambda(\mathbf{k}) W^*_{\lambda'}(\mathbf{k}') \rangle = \delta_{\lambda\lambda'} \delta^3(\mathbf{k} - \mathbf{k}')$:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \sum_\lambda \int d^3k \, |\hat{\epsilon}_\lambda|^2 \frac{\hbar\omega}{8\pi^2} \tag{10}$$

Since $\sum_{\lambda=1}^2 |\hat{\epsilon}_\lambda|^2 = 2$ (two orthogonal unit polarization vectors):

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \int d^3k \, \frac{\hbar\omega}{4\pi^2} \tag{11}$$

Therefore:

$$\langle \mathbf{E}^2 \rangle = 2 \int d^3k \, \frac{\hbar\omega}{4\pi^2} \tag{12}$$

By identical calculation:

$$\langle \mathbf{B}^2 \rangle = 2 \int d^3k \, \frac{\hbar\omega}{4\pi^2} \tag{13}$$

The total energy density is:

$$\langle u \rangle = \frac{1}{8\pi}(\langle \mathbf{E}^2 \rangle + \langle \mathbf{B}^2 \rangle) = \frac{1}{8\pi} \cdot 4 \int d^3k \, \frac{\hbar\omega}{4\pi^2} = \frac{1}{(2\pi)^3} \int d^3k \, \hbar\omega \tag{14}$$

Since $d^3k/(2\pi)^3$ represents the density of states in momentum space (the number of modes per unit volume in k-space), we can interpret this as:

$$\langle u \rangle = \int \frac{d^3k}{(2\pi)^3} \, \hbar\omega = \int [\text{modes per } k^3] \times [\text{energy per mode}] \tag{15}$$

This confirms that each electromagnetic normal mode (specified by wavevector $\mathbf{k}$ and polarization λ) carries energy $(1/2)\hbar\omega$ in the zero-point radiation field:

$$\boxed{\rho(\omega) = \frac{1}{2}\hbar\omega \quad [\text{energy per mode}]} \tag{16}$$

Comment: I have a note here, that like basically maybe I can delete the above two equations ... but will decide on the next editing job.

4 Maxwell Equations for Gravity

In this section we introduce the second of our two assumptions. We call this an assumption because Heaviside first proposed this a long time ago (Clearly need some reference here and help writing that sentence better). We assume that gravity obeys the same set of equations as that that governs electromagntism, up to a numeric factor in the coupling constant. While the Maxwell equations were originally written for charge and charge currents, we replace these with mass and mass currents acting as sources. In natural units ($c = 1$):

$$\begin{aligned} \nabla \times \mathbf{B} - \partial_t \mathbf{E} &= -4\pi G \mathbf{J}, & \nabla \times \mathbf{E} + \partial_t \mathbf{B} &= 0 \\ \nabla \cdot \mathbf{E} &= -4\pi G \rho, & \nabla \cdot \mathbf{B} &= 0 \end{aligned} \tag{17}$$

These equations describe a vector field sourced by mass density ρ and mass current $\mathbf{J}$, with G the gravitational coupling constant, 6.67 something (Next edit fix this).

Some may be surprised by the presence of $\mathbf{B}$, a magnetic-like field. But this emerges naturally: just as a moving charge generates a magnetic field, a moving mass generates a "gravitomagnetic" field. While static masses generate only $\mathbf{E}$, moving masses induce $\mathbf{B}$; however, the effect is negligible unless the mass is both extremely large and rapidly moving. **Section 7** discusses a physical manifestation of this effect and explores its potential testability.

Much of what follows is well-known but is necessary mathematical tools and techniques to discuss gravity and radiation in subsequent sections. (Awkward writing, Claude, please help.)

4.1 Irrotational and Solenoidal Decomposition

Any vector field can be split into irrotational (curl-free) and solenoidal (divergence-free) components:

$$\mathbf{V} = \mathbf{V}_I + \mathbf{V}_S, \quad \boldsymbol{\nabla} \cdot \mathbf{V}_S = 0, \quad \boldsymbol{\nabla} \times \mathbf{V}_I = 0 \tag{18}$$

Applying this to our gravitational fields yields two decoupled subsystems:

Static Fields (Irrotational):

$$\boldsymbol{\nabla} \cdot \mathbf{E}_I = -4\pi G\rho, \quad \partial_t \mathbf{E}_I = 4\pi G \mathbf{J}_I, \quad \boldsymbol{\nabla} \cdot \mathbf{J}_I + \partial_t \rho = 0 \tag{19}$$

Radiation Fields (Solenoidal):

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 4\pi G \mathbf{J}_S, \quad \boldsymbol{\nabla} \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0 \tag{20}$$

This decomposition reflects a meaningful physical distinction between the **static gravitational attraction** and the **propagating radiation** field. Though derivable from the Maxwell equations, we list the continuity equation separately as it plays a direct role in what follows.

In the rest frame of the mass, these sectors decouple. One set of equations governs the gravitational field sourced by the mass, while another describes the radiation field. This separation enables a detailed analysis of radiation without affecting the static component. ("Affecting" is the wrong word here?)

We can make this explicit by defining a new current:

$$\mathbf{J}'_{\mathbf{S}} = -G\mathbf{J}_{\mathbf{S}} \tag{21}$$

This transforms the inhomogeneous solenoidal equation into:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 4\pi \mathbf{J}'_{\mathbf{S}} \tag{22}$$

while leaving all other equations unchanged.

Since this form matches the standard presentation in textbooks on electromagnetic radiation, we will drop the prime and adopt the following as the solenoidal (radiation) Maxwell equations:

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S &= 4\pi \mathbf{J}_S \\ \boldsymbol{\nabla} \times \mathbf{E}_S + \partial_t \mathbf{B}_S &= 0 \end{aligned} \tag{23}$$

4.2 Spherical Vector Harmonics

The following sections derive the explicit solutions to the Maxwell equations used in this theory.

Due to the symmetric radial symmetry of the problem, we will benefit from using Vector Spherical Harmonics.

Following Berrara (http://iopscience.iop.org/article/10.1088/0143-0807/6/4/014/meta) (we should send this to the references when we get a chance) we assume that any vector field, $\mathbf{V}$, and any scalar, g, can be expanded as follows:

$$\mathbf{V}(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \big[a_{lm}(r)\,\mathbf{Y}_{lm}(\theta,\phi) + b_{lm}(r)\,\hat{\mathbf{r}} \times \mathbf{X}_{lm}(\theta,\phi) + c_{lm}(r)\,\mathbf{X}_{lm}(\theta,\phi)\big] \tag{24}$$

$$g(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} g_{lm}(r) Y_{lm}(\theta,\phi) \tag{25}$$

where Y_{lm} are the usual spherical harmonics (see Jackson 3.53), $\mathbf{Y}_{lm}$ is simply $\hat{\mathbf{r}}Y_{lm}$ and $\mathbf{X}_{lm}$ is defined as,

$$\mathbf{X}_{lm}(\theta,\phi) = \frac{-i}{\sqrt{l(l+1)}} \mathbf{r} \times (\boldsymbol{\nabla} Y_{lm}(\theta,\phi)) \tag{26}$$

4.3 Recovery of Newtonian Gravity (Irrotational Solutions)

We now calculate Newtonian gravity as a solution to the Irrotational parts of the Maxwell equations.

In the case of a static point mass m located at the origin, (We will have much more to discuss about this in possibly eqn 52 or 34 ... but they've probably changed R = Gm equation and sigma = m / h delta) (we have question marks over here???) we set:

$$\rho(\mathbf{x}) = m\delta^3(\mathbf{x}), \quad \mathbf{J} = 0 \tag{27}$$

Using the irrotational part of the Maxwell equations:

$$\boldsymbol{\nabla} \cdot \mathbf{E}_I = -4\pi G\rho \qquad \boldsymbol{\nabla} \times \mathbf{E}_I = 0 \tag{28}$$

and using the spherical vector harmonics we developed earlier, we find:

$$\mathbf{E}_I(\mathbf{x}) = \sum_{l,m} \left[-\frac{4\pi i G}{2l+1} \frac{q_{lm}}{r^{l+2}} \sqrt{l(l+1)}\,(\hat{\mathbf{r}} \times \mathbf{X}_{lm}) - \frac{4\pi G(l+1)}{2l+1} \frac{q_{lm}}{r^{l+2}} \mathbf{Y}_{lm} \right] \tag{29}$$

Where,

$$q_{lm} \equiv \int Y^*_{lm}(\theta,\phi) r^l \rho(\mathbf{x})\; d^3x = \int \rho_{lm}(r) r^{l+2}\; dr \tag{30}$$

Clearly then, if

$$\rho(\mathbf{x}) = m\delta^3(\mathbf{x}) \tag{31}$$

We get

$$\mathbf{E}_I(\mathbf{x}) = -\hat{\mathbf{r}} \frac{mG}{r^2} \tag{32}$$

as expected.

4.4 Field Momentum in SED

4.5 Radiation from the Solenoidal Field (Solenoidal Solutions)

Several results from the solutions of the solenoidal Maxwell equations will be needed later; we now solve the corresponding equations.

Outside the mass, the solenoidal field equations in vacuum are:

$$\nabla \times \mathbf{E}_S + \partial_t \mathbf{B}_S = 0, \quad \nabla \times \mathbf{B}_S - \partial_t \mathbf{E}_S = 0 \tag{33}$$

Assuming time dependence $e^{-i\omega t}$, the general solution is expanded in vector spherical harmonics:

$$\begin{aligned} \mathbf{B}_S &= \sum_{l,m} \left[a_E(l,m) f_l(kr) \mathbf{X}_{lm} - \frac{i}{k} a_M(l,m) \nabla \times (g_l(kr)\mathbf{X}_{lm}) \right] \\ \mathbf{E}_S &= \sum_{l,m} \left[\frac{i}{k} a_E(l,m) \nabla \times (f_l(kr)\mathbf{X}_{lm}) + a_M(l,m) g_l(kr) \mathbf{X}_{lm} \right] \end{aligned} \tag{34}$$

where $\mathbf{X}_{lm}$ was defined in (26).

The functions f_l and g_l are linear superpositions of the radial Hankel functions,

$$h_l^{(1)}(x) = -i(-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \left(\frac{e^{ix}}{x} \right) \tag{35}$$

$$h_l^{(2)}(x) = \left(h_l^{(1)}(x) \right)^* \tag{36}$$

a_E and a_M as well as the particular dependence of f_l and g_l on $h_l^{(1)}(x)$ and $h_l^{(2)}(x)$ depend on the boundary conditions.

In our case, we are looking at a massive particle that sinks radiation, we therefore need to only include incoming waves.

The total instantaneous power absorbed is defined as.

$$P \equiv \int_V d^3x \; \nabla \cdot \left(\frac{1}{8\pi} \mathbf{E} \times \mathbf{B}^* \right) = \oint_{\partial V} \hat{\mathbf{r}} \cdot \mathbf{S} \; da = R^2 \int d\Omega \; \hat{\mathbf{r}} \cdot \mathbf{S} \Big|_{r=R} \tag{37}$$

Using the relationships in the introduction on Vector Spherical harmonics, we find that we are left with only two non-vanishing terms,

$$P = \frac{R^2}{8\pi} \sum_{l,m} \left[\frac{-i}{k} |a_E|^2 h_l^{(2)} \frac{1}{r} \frac{d}{dr} \left(r h_l^{(1)} \right) + \frac{i}{k} |a_M|^2 h_l^{(1)} \frac{1}{r} \frac{d}{dr} \left(r h_l^{(2)} \right) \right]_{r=R} \tag{38}$$

Since, by assumption (See Section 2.1) the massive particle is electromagnetically neutral on average, we need neither the Transverse Magnetic nor the

Transverse Electric modes to dominate. Thus, the two proportionality terms must obey $|a_E|^2 = |a_M|^2$. Use of the Wronskian leaves a particularly simple form for the power radiated,

$$P = \sum_{l,m} \frac{|a_E|^2}{4\pi k^2} \tag{39}$$

The associated momentum density also follow cleanly. The total field momentum is directed radially inward:

$$\mathbf{P}_{\text{field}} = \frac{1}{8\pi} \int_V \mathbf{E} \times \mathbf{B}^* \, d^3x = -\hat{\mathbf{r}} \sum_{l,m} \int_V \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{4\pi k^2 r^2} \, d^3x \tag{40}$$

As mentioned in subsection earlier, Both approaches lead to $\mathbf{E} \times \mathbf{B}^*$ appearing in the final expressions, but for different physical reasons.

Comment: obviously, find the subsection earlier (when we introduce ZPR)

4.6 hopefully the editing flowed

4.6.1 Full Field Expansion with Stochastic Coefficients

We write the complete multipole expansion including both electric and magnetic type modes:

$$\begin{aligned}
\mathbf{B} &= \int d\omega \sum_{l,m} \left[a_E(l,m) f_l(kr) \mathbf{X}_{lm} - \frac{i}{k} a_M(l,m) \boldsymbol{\nabla} \times (g_l(kr) \mathbf{X}_{lm}) \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.} \\
\mathbf{E} &= \int d\omega \sum_{l,m} \left[\frac{i}{k} a_E(l,m) \boldsymbol{\nabla} \times (f_l(kr) \mathbf{X}_{lm}) + a_M(l,m) g_l(kr) \mathbf{X}_{lm} \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.}
\end{aligned} \tag{41}$$

where the functions f_l and g_l are linear superpositions of radial Hankel functions $h_l^{(1)}(kr)$ and $h_l^{(2)}(kr)$, and $W_{l,m}(\omega)$ are complex Gaussian stochastic variables with:

$$\langle W_{l,m}(\omega) \rangle = 0, \quad \langle W_{l,m}(\omega) W^*_{l',m'}(\omega') \rangle = \delta_{ll'} \delta_{mm'} \delta(\omega - \omega') \tag{42}$$

For a massive particle that absorbs radiation (only incoming waves), we have:

$$f_l(kr) = h_l^{(1)}(kr), \quad g_l(kr) = h_l^{(1)}(kr) \tag{43}$$

4.6.2 Computing the Field Momentum

The total field momentum is:

$$\mathbf{P}_{\text{field}} = \frac{1}{4\pi} \int_V \mathbf{E} \times \mathbf{B} \, d^3x \tag{44}$$

Comment: might be an 8 in that denominator

Following the math we laid out in Section ?, we compute the stochastic average $\langle \mathbf{E} \times \mathbf{B} \rangle$. Writing $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ and $\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*$, the terms that survive the stochastic average are those involving $\langle WW^* \rangle$:

$$\langle \mathbf{E} \times \mathbf{B} \rangle = \langle \mathbf{E}_+ \times \mathbf{B}_+^* \rangle + \langle \mathbf{E}_+^* \times \mathbf{B}_+ \rangle \tag{45}$$

Working through the cross products using the orthogonality relations of vector spherical harmonics, and noting that $h_l^{(1)*} = h_l^{(2)}$, the angular integrals over $|\mathbf{X}_{lm}|^2$ yield normalization constants that can be absorbed.

$$\langle \mathbf{P}_{\text{field}} \rangle = -\hat{\mathbf{r}} \int dr \int d\omega \sum_{l,m} \left[\frac{-i}{k} |a_E|^2 h_l^{(2)} \frac{1}{r} \frac{d}{dr} \left(r h_l^{(1)} \right) + \frac{i}{k} |a_M|^2 h_l^{(1)} \frac{1}{r} \frac{d}{dr} \left(r h_l^{(2)} \right) \right] \tag{46}$$

The above can be simplified by use of the Wronskian, leaving:

$$\boxed{\langle \mathbf{P}_{\text{field}} \rangle = -\hat{\mathbf{r}} \int dr \int d\omega \sum_{l,m} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{4\pi k^2 r^2}} \tag{47}$$

4.6.3 Field Energy in the Radiation Zone

The total field energy density is given by:

$$u = \frac{1}{8\pi} (\mathbf{E}^2 + \mathbf{B}^2) \tag{48}$$

Computing the stochastically averaged energy density for the full field is complicated due to cross terms between electric and magnetic multipoles. However, in the radiation zone where $k^2 r^2 >> 1$, the fields simplify considerably.

In the radiation zone, using $f_l(kr) \approx h_l^{(1)}(kr) \approx \frac{(-i)^{l+1}}{kr} e^{ikr}$ for incoming waves, the stochastic fields become:

$$\begin{aligned}
\mathbf{B} &= \frac{e^{ikr}}{kr} \int d\omega \sum_{l,m} (-i)^{l+1} \left[a_E(l,m) \mathbf{X}_{lm} + a_M(l,m) \hat{\mathbf{r}} \times \mathbf{X}_{lm} \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.} \\
\mathbf{E} &= \frac{e^{ikr}}{kr} \int d\omega \sum_{l,m} (-i)^{l+1} \left[a_M(l,m) \mathbf{X}_{lm} - a_E(l,m) \hat{\mathbf{r}} \times \mathbf{X}_{lm} \right] e^{-i\omega t} W_{l,m}(\omega) + \text{c.c.}
\end{aligned} \tag{49}$$

Writing $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*$ and $\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*$, we compute:

$$\langle \mathbf{E}^2 \rangle = 2 \langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle, \quad \langle \mathbf{B}^2 \rangle = 2 \langle \mathbf{B}_+ \cdot \mathbf{B}_+^* \rangle \tag{50}$$

For $\mathbf{E}_+ \cdot \mathbf{E}_+^*$:

$$\langle \mathbf{E}_+ \cdot \mathbf{E}_+^* \rangle = \frac{1}{k^2 r^2} \int d\omega \sum_{l,m} \left[|a_M|^2 |\mathbf{X}_{lm}|^2 + |a_E|^2 |\hat{\mathbf{r}} \times \mathbf{X}_{lm}|^2 \right] \tag{51}$$

where the cross terms vanish due to the orthogonality $\mathbf{X}_{lm} \cdot (\hat{\mathbf{r}} \times \mathbf{X}_{lm}) = 0$. Similarly:

$$\langle \mathbf{B}_+ \cdot \mathbf{B}_+^* \rangle = \frac{1}{k^2r^2} \int d\omega \sum_{l,m} \left[|a_E|^2 |\mathbf{X}_{lm}|^2 + |a_M|^2 |\hat{\mathbf{r}} \times \mathbf{X}_{lm}|^2 \right] \tag{52}$$

Since $|\hat{\mathbf{r}} \times \mathbf{X}_{lm}| = |\mathbf{X}_{lm}|$ (both are unit vectors perpendicular to $\hat{\mathbf{r}}$), and using the neutrality condition $|a_E|^2 = |a_M|^2$ (see Section 2.1), we obtain:

$$\langle \mathbf{E}^2 + \mathbf{B}^2 \rangle = \frac{4}{k^2r^2} \int d\omega \sum_{l,m} |a_E|^2 |\mathbf{X}_{lm}|^2 \tag{53}$$

The total field energy is therefore:

$$\boxed{\langle U \rangle = \int_V d^3x \int d\omega \sum_{l,m} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{4\pi k^2 r^2}} \tag{54}$$

The expansion of this calculation outside the radiation zone involves more intricate mathematics due to the full radial dependence of the Hankel functions and non-vanishing cross terms between multipoles. We defer this more general treatment to future work.

4.7 Confused about where we are

The above was calculated exactly. However, the total field energy is more difficult to calculate. In the present work we only calculte the energy in the radiation zone: $k^2r^2 >> 1$. We discuss aspects of this further in **??**. In the radiation zone, we can use the approximation:

$$\begin{aligned} \mathbf{B}_S &= \frac{e^{-ikr}}{kr} \sum_{l,m} (-i)^{l+1} \left[a_E(l,m) \mathbf{X}_{lm} + a_M(l,m) \hat{\mathbf{r}} \times \mathbf{X}_{lm} \right] \\ \mathbf{E}_S &= \frac{e^{-ikr}}{kr} \sum_{l,m} (-i)^{l+1} \left[a_M(l,m) \mathbf{X}_{lm} - a_E(l,m) \hat{\mathbf{r}} \times \mathbf{X}_{lm} \right] \end{aligned} \tag{55}$$

From this it follows straightforwardly that

$$U = \sum_{l,m} \int_V \frac{1}{4\pi} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{k^2 r^2} \, d^3x \tag{56}$$

The expansion of this outside of the radiation zone leads to intriguing math but we must save it for future research .

The specific coefficients $|a_E|^2$ will be fixed by boundary conditions in the section 6 on collapse and stability. But, since we've already spelled out the hypothesis several times now, then we know that we will eventually land on :

$$\boxed{\mathbf{p}(r) = -\hat{\mathbf{r}} \frac{Gm\hbar}{r^2}} \tag{1}$$

This result follows from boundary conditions at the mass's surface.

Comment: I think we will have to print out this version to see what we messed up ... because things are in crazy places ... hope I didn't erase anything important.

Comment: Shoot and the git history was no good. So I need to rely on the paper version if I erased anything important ... I don't think I did though.

5 A Toy Model

In this section we introduce a module of a massive particle. To gain traction on the problem, we must make several simplifying assumptions .

First, in talking about the space domain, we will quote from Rohrlich at length to find a lower bound for the radius. We will therefore restrict our attention to $R \geq r$ and assume a shell ... mainly because we will promise ourselves to never even consider radii smaller than the limit found in the next section.

After tha, we will move to the frequency domain, and we will speak about the frequency of radiation absorbed by the massive particle (via our hypothesis) and show that for a simple model that only absorbs one frequncy spectra, that we are guaranteed that the massive particle absorbs less energy than is available in the stochastic zero-point radiation field.

5.1 Lower Bound 3: Minimum Radius via Gravitational Self-Energy

The following long quote is from Rohrlich because I can't say it any better than he did. It has been modified from the original to reflect that we are talking about massive particles and not charged particles.

We wish to construct a model of a massive particle. "The most obvious model of a massive particle is a sphere carrying a spherically symmetrical mass distribution. While such a model is meant (and was indeed proposed) as a picture of a massive elementary particle (a neutron, for example), it is obvious that is is basically a macroscopic massive body, only much smaller. There is nothing 'elementary' about it. To see this clearly, let us consided a macroscopic massive sphere in more detail.

"Consider a sphere of radius R and spherically symmetric mass density $\rho(\mathbf{x})$. Its total mass is

$$m = \int_V \rho(\mathbf{x}) d^3x \tag{57}$$

"The integral extending over the volume of the sphere. Since $m \neq 0$ by assumption, there must exist attractive gravitational forces between 'parts' of the total mass. Symmetry would therefore require that this massive sphere contract and continue to do so to arbitrarily small radius R." Thus with no other forces present, the particle will necessaruly collapse.

In the next section, I will show that the particle need not collapse. For now, however, let us take it as an assumption that the particle does not collapse.

Since we are assuming that there are no other forces in play besides for the gravitational one, its total energy must be

$$U = m = -\frac{1}{2}\int \frac{Gm\rho(\mathbf{x})}{r} d^3x \tag{58}$$

We note that the more fundamental relativistic expression is $U^2 = m^2$ (or equivalently, $p_\mu p^\mu = m^2$ in four-vector notation), which we utilize and take

$$U^2 = m^2 \tag{59}$$

To complete the computation, we must choose some mass density distribution to integrate over. Mathematically, the simplest is a uniform spherical shell at the radius R, in which case

$$m = \frac{Gm^2}{2R} \qquad or \qquad R = \frac{Gm}{2} \tag{60}$$

If, however, we choose a different mass density distribution (a uniform sphere for example), we find that the radius R will always be of the order Gm. Therefore, I have chosen $R = Gm$ as the mathematical limit of the radius of a massive particle. This radius is similar in spirit to the classical radius of an electron.

It is useful to compare the mathematical radius with the radii of real physical bodies. Slipping back to SI units, we give a short list of real macroscopice objects with $\Phi = \frac{Gm}{Rc^2}$

$$\text{Mathematical limit:} \quad \Phi = 1 = \frac{Gm}{Rc^2}$$

$$\text{Black hole:} \quad \Phi = 1$$

$$\text{Sun:} \quad \Phi = 2 \times 10^{-6}$$

$$\text{Earth:} \quad \Phi = 7 \times 10^{-10}$$

$$\text{Proton:} \quad \Phi = 1 \times 10^{-38}$$

5.2 Frequency Stuff for our model

It should be clear that any real model of a massive object that absorbs radiation will absorb radiation over a non-singular domain. However, for ease of .... stuff, to gain traction. We will pick a model that only absorbs one particular frequency. Unsurprisingly, we choose the frequency that is absorbed by our massive particle to be

$$\omega = \frac{m}{\hbar} \tag{61}$$

Or, re-writing this as a delta function, in spherical coordinates gives:

$$\frac{\delta\left(\omega - \frac{m}{\hbar}\right)}{4\pi\omega^2} \tag{62}$$

We can then compute the energy that is absorbed from the field

Comment: (I'm assuming we have an equation that lines up with this in the stochastic section)

$$U = \int d\omega \quad \hbar\omega\delta\left(\omega - \frac{m}{\hbar}\right) \tag{63}$$

Which yields

$$U = m \tag{64}$$

To show that it draws less energy than is available, we will add a constant multiplicative factor α and re-write the above analysis

$$U = m = \frac{Gm^2}{R} \tag{65}$$

and

$$U = \alpha \int \hbar\omega\delta\left(\omega - \frac{m}{\hbar}\right) \quad d\omega \tag{66}$$

This yields

$$\alpha = \frac{Gm}{R} \tag{67}$$

therefore if $R > Gm$ then $\alpha < 1$ and we draw less energy from the Z.P.R. than is available.

6 Field Energy and Momentum Balance: A Mechanism for Stability

In this section, we examine whether the proposed inflow of zero-point radiation can stabilize a massive particle against gravitational collapse. If the energy absorbed from the vacuum field offsets the divergent self-energy of the particle's own field, a natural mechanism for stability emerges. We first analyze the energy density and total energy of the system, incorporating a high-frequency cutoff introduced earlier. We then explore whether this energy balance leads to a self-consistent equilibrium. Finally, we turn to momentum conservation, where we derive an effective "Ohm's law" relating solenoidal fields and currents. Together, these results show that the hypothesized momentum inflow can provide a classical mechanism of stability rooted in the structure of the zero-point field itself.

As noted in the previous section **Section 5** we need to use a toy model. And we will choose the spherical shell. So that, if we ever say "on shell" it will be clear.

Comment: We should mention that we will be using the "toy" model from the previous section and threfore restricting our attention to eht surface of the particle or being "on the shell"

6.1 Stability from Field Energy

We now evaluate whether the inward radiation flow proposed in **Section 2** can provide a stabilizing mechanism for a massive particle. Specifically, we ask: can the absorbed energy from the zero-point field counterbalance the gravitational self-energy that would otherwise cause collapse?

The total field energy is given by:

$$U = \frac{1}{16\pi} \int \left(|\mathbf{E}|^2 + |\mathbf{B}|^2 \right) \, d^3x \tag{68}$$

If we pause and look at just the irrotational part of hese equations. we have

$$U_I = \frac{1}{16\pi} \int \left(|\mathbf{E_I}|^2 \right) \, d^3x \tag{69}$$

This leads to

$$U_I \sim \int \left(\frac{Gm^2}{r^4} \right) \, d^3x \tag{70}$$

Which is clearly divergent and therefore a mass would collapse under its own weight without some kind of counter balance force thing.

Comment: Yeah ... I totally used my memory for the above equation, so we should double check it in the next pass

Unsurprisingly, the hypothesis of this paper is that we should include the radiation contribution, and thus we have:

The total energy, combining both irrotational and solenoidal contributions, is:

$$U_{\text{tot}} = \int_V u \, d^3x = \int_V \left(-\frac{1}{4\pi G} |\mathbf{E}_I|^2 + |\mathbf{E}_S|^2 + |\mathbf{B}_S|^2 \right) \, d^3x \tag{71}$$

Here of course we are "on shell".

From equation (56), we have:

$$U_{\text{rad}} = \sum_{l,m} \int_V \frac{1}{4\pi} \frac{|a_E|^2 |\mathbf{X}_{lm}|^2}{k^2 r^2} \, d^3x \tag{72}$$

The above is of course us being very careful about averaging over the stochastic noise as presented previously (hopefully have an equation to point to)

And naturally using our hypothesis and the arguments in section **??** (above about ... getting tired) we then have

$$U_{\text{rad}} = \frac{1}{8\pi^3} \int \int \frac{Gm\hbar}{r^2} \, d^3x \, d^3k \tag{73}$$

Integrating over k and combining, gives the following total energy:

$$U_{\text{tot}} = \int \left[\frac{Gm\hbar}{r^2} k^3 - \frac{Gm^2}{r^4} \right] \, d^3x \tag{74}$$

Comment: Using the heuristics we developed in subsection ??, that were admittedly slightly fishy but we had decided were not fatal to the theory, namely that

This is no longer anything. This is a super straightforward algebra problem, assuming we are on shell, so that $k = m/\hbar$ and with $r = R$

$$k_{\max} = \frac{m}{\hbar} \qquad \textit{and} \quad r^2k^2 = 1 \tag{75}$$

And of course just as we had "numerical prefactors" in section about mass r = R, here we do suppress numerical prefactors and absorb them into the definition of the model.

we then see that the energy is at most a constant and therefor is conseved

6.2 Momentum Conservation and "Ohm's Law"

We now turn to momentum conservation. The total momentum exchange in the system should obey:

$$\int_V (\rho \mathbf{E}_I + \mathbf{J}_S \times \mathbf{B}_S)\, d^3x = 0 \tag{76}$$

Assuming the solenoidal current obeys a linear response:

$$\mathbf{J}_S = \sigma \mathbf{E}_S \tag{77}$$

And assuming that we use our model, such that

$$\rho = m\delta^3(\mathbf{x}) \tag{78}$$

is actually a call to use our model and be "on shell", such that we do not actually try to integrate over spheres smaller than the smallest radius that was outlind ine **Section ?**

and harmonic time dependence $e^{-i\omega t}$, and then averaging over the stochastic noise the field momentum becomes:

$$-\hat{\mathbf{r}} \int \left(m\delta^3(\mathbf{x}) \cdot \frac{Gm}{r^2} - \sigma \cdot \frac{Gm\hbar}{r^2} \right) d^3x = 0 \tag{79}$$

Solving gives:

$$\sigma = \frac{m}{\hbar}\delta^3(\mathbf{x}) \tag{80}$$

This expression defines a delta-function conductivity concentrated at the origin, but knowing we actually mean the shell. Thus this is an effective Ohm's law in which the mass acts as a point-like, perfect absorber for incoming radiation as long as we are outside of the radius in **Section ?**. We can re-write the relevant Maxwell equation in light of this:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar}\mathbf{E}_S\delta^3(\mathbf{x}) \tag{81}$$

Together, the field energy and momentum conservation arguments demonstrate that the hypothesized momentum inflow can act as a self-consistent stabilizing structure. A massive particle that sinks zero-point radiation does not collapse because of the interaction with the vacuum field itself.

7 Classical Tests of Gravity

This section demonstrates that the present theory reproduces three key experimental predictions of general relativity: the precession of Mercury's perihelion, the deflection of light near a massive object, and gravitational redshift.

Comment: These are the three classic tests of general relativity ... like pretty sure that Einstein himself propsed these ... or at minimum ... like they're really famous ... we should double check this, and then also mention this up above in the intro and the conclusion

7.1 Precession of the Perihelion of Mercury

In electrodynamics, a static Coulomb field transforms under Lorentz boosts into a configuration with both electric and magnetic components. Specifically, the fields of a charge moving at velocity $\mathbf{v}$ are:

$$\mathbf{E}' = \gamma\mathbf{E} - \frac{\gamma^2}{\gamma+1}\mathbf{v}(\mathbf{v}\cdot\mathbf{E}), \quad \mathbf{B}' = -\gamma\mathbf{v}\times\mathbf{E} \tag{82}$$

where $\gamma = 1/\sqrt{1-v^2}$.

By analogy, we assume the same Lorentz transformation laws apply to gravity. We define the gravitational field:

$$\mathbf{E}_g = \mathbf{F}/m = -\frac{GM}{r^3}\mathbf{r} \tag{83}$$

Then, under a Lorentz boost:

$$\mathbf{E}_g' = \gamma\mathbf{E}_g - \frac{\gamma^2}{\gamma+1}\mathbf{v}(\mathbf{v}\cdot\mathbf{E}_g), \quad \mathbf{B}_g' = -\gamma\mathbf{v}\times\mathbf{E}_g \tag{84}$$

In the low-velocity limit $v \ll 1$, we find:

$$\mathbf{B}_g = -\mathbf{v}\times\mathbf{r}\cdot\frac{GM}{r^3} \tag{85}$$

This introduces a "gravitomagnetic" interaction analogous to magnetism in electrodynamics.

We now consider Kepler's problem. The Newtonian effective potential is:

$$U_{\text{Newt}} = \frac{1}{2}m\left(\frac{dr}{dt}\right)^2 - \frac{GMm}{r} + \frac{l^2}{2mr^2} \tag{86}$$

where $l = m|\mathbf{r}\times\mathbf{v}|$ is the orbital angular momentum (see Figure 1).

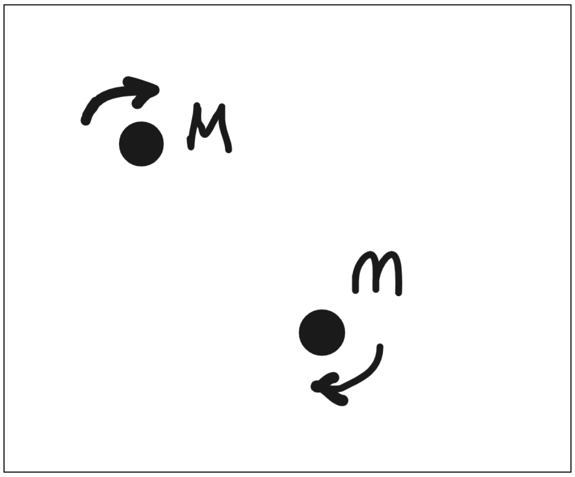

Figure 1: Two masses in mutual orbit

A moving charge induces a magnetic moment $\mathbf{m} = \frac{1}{2}e(\mathbf{r} \times \mathbf{v})$. Replacing $e \to m$, we define a gravitational magnetic moment:

$$\mathbf{m}_g = \frac{1}{2}m(\mathbf{r} \times \mathbf{v}) \tag{87}$$

The potential energy of this moment in a magnetic field is:

$$U = -\mathbf{m}_g \cdot \mathbf{B}_g = -\frac{GMm}{2}|\mathbf{r} \times \mathbf{v}|^2/r^3 \tag{88}$$

This adds a correction to the potential energy. Including the reciprocal effect (from the motion of the central mass), we double this term and find the corrected potential:

$$U_{\text{Einstein}} = \frac{1}{2}m\left(\frac{dr}{dt}\right)^2 - \frac{GMm}{r} + \frac{l^2}{2mr^2} - \frac{GMl^2}{mr^3} \tag{89}$$

This is the same correction obtained in general relativity to first order. As Carroll notes, general relativity produces this result exactly; in our theory, it emerges as a first-order approximation. Higher-order terms may offer a route to testing deviations.

7.2 Deflection of Light

We now consider the deflection of light near a massive object. Our central hypothesis is that mass creates a momentum-transfer field:

$$\mathbf{P}(r) = -\hat{\mathbf{r}}\frac{Gm\hbar}{r^2} \tag{1}$$

A photon traveling through this field receives a cumulative momentum shift:

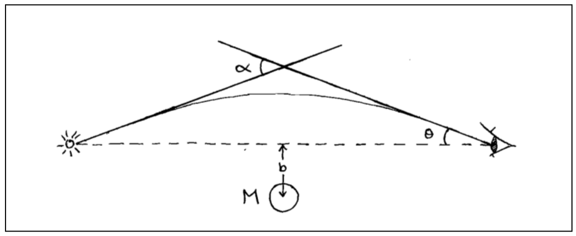

Figure 2: Deflection of light

$$\Delta\mathbf{p} = \int_{\text{path}} \mathbf{P}(r)\, dl \tag{2}$$

Assuming a straight-line path passing at minimum distance b (see figure 2), we integrate:

$$\Delta\mathbf{p} = \int_{-\infty}^{\infty} -\frac{Gm\hbar}{r^3}\mathbf{r}\, w\, dz \tag{90}$$

The parallel momentum shifts cancel symmetrically. The transverse component yields:

$$\Delta p_y = \hbar w \cdot \frac{2Gm}{b} \tag{91}$$

The angle of deflection is then:

$$\theta \approx \frac{\Delta p_y}{p_z} = \frac{2Gm}{b} \quad \Rightarrow \quad \alpha = \frac{4Gm}{b} \tag{92}$$

This matches, to first-order, the deflection angle predicted by general relativity.

7.3 Gravitational Redshift

Finally, we consider the redshift of light climbing out of a gravitational well. As before, a photon gains or loses momentum depending on its path through the momentum field.

Using:

$$\Delta p = \int_{\infty}^{R} \mathbf{P} \cdot \hat{\mathbf{r}} \cdot w\, dr = \int_{\infty}^{R} -\frac{Gm\hbar}{r^2} w\, dr = \hbar w \cdot \frac{Gm}{R} \tag{93}$$

The resulting frequency shift is:

$$\hbar w' = \hbar w + \Delta p = \hbar w \left(1 + \frac{Gm}{R}\right) \tag{94}$$

which, to first order, matches the prediction of general relativity for gravitational redshift.

8 Duality and the Dirac Equation

Throughout this paper, we've stayed grounded in classical physics. We built a theory of gravity using field equations, conservation laws, and an underlying sea of vacuum radiation: no quantum assumptions, no curved spacetime. In this section, something surprising appears. When the field equations are rewritten in a new form, a familiar structure emerges: the Dirac equation, the cornerstone of quantum mechanics.

8.1 Duality Transformations

As discussed in Jackson (Section 6.11), the Maxwell equations can be extended to include both electric and magnetic charges:

$$\begin{aligned} \nabla \times \mathbf{B} - \partial_t \mathbf{E} &= \mathbf{J}_e, \qquad & \nabla \times \mathbf{E} + \partial_t \mathbf{B} &= -\mathbf{J}_m \\ \nabla \cdot \mathbf{E} &= \rho_e, & \nabla \cdot \mathbf{B} &= \rho_m \end{aligned} \tag{95}$$

The magnetic densities are assumed to obey the same continuity equation as their electric counterparts.

Now consider the following duality transformation:

$$\begin{aligned} \mathbf{E} &= \mathbf{E}' \cos\xi + \mathbf{B}' \sin\xi, \qquad & \mathbf{B} &= \mathbf{B}' \cos\xi - \mathbf{E}' \sin\xi \\ \rho_e &= \rho_e' \cos\xi + \rho_m' \sin\xi, & \rho_m &= \rho_m' \cos\xi - \rho_e' \sin\xi \\ \mathbf{J}_e &= \mathbf{J}_e' \cos\xi + \mathbf{J}_m' \sin\xi, & \mathbf{J}_m &= \mathbf{J}_m' \cos\xi - \mathbf{J}_e' \sin\xi \end{aligned} \tag{96}$$

This transformation leaves the generalized Maxwell equations invariant. As Jackson notes, the distinction between electric and magnetic charge is, in this framework, a matter of convention. If all particles share the same magnetic-to-electric charge ratio, a duality rotation can eliminate magnetic charges entirely. This perspective opens the door for deeper reinterpretations of classical field theory.

8.2 Spinor Representation of Field Operators

We now examine a way to recast Maxwell's equations using spinor-like objects.

Define the vector curl as a matrix operator:

$$\nabla \times \mathbf{V} = g_i \partial_i \mathbf{V} \tag{97}$$

with:

$$g_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad g_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{98}$$

Define the gamma matrices as (These matrices satisfy the algebra of infinitesimal rotations and generate the curl operator in 3D"):

$$\gamma^0 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & g_i \\ -g_i & 0 \end{pmatrix} \tag{99}$$

These matrices are not the standard Dirac representation, but they serve to connect the structure of the field equations to the spinor formalism in a suggestive way. This could be further explored with tools from representation theory and Clifford algebras.

8.3 Emergence of the Dirac Equation

Recall from the section on momentum conservation that we have quation (78):

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar}\mathbf{E}_S \delta^3(\mathbf{x}) \tag{78}$$

Restricting attention to the surface of the particle (i.e., “on-shell”). To understand the ”on-shell” approximation, we must clarify our model of the massive particle. Rather than treating mass as a point source, we model it as a thin spherical shell of radius $R \sim Gm$, consistent with the spatial cutoff discussed in Section 5. The mass density is thus $\rho = (m/4\pi R^2)\delta(r - R))$ When we say we are working ”on-shell,” we mean restricting our attention to the surface $r = R$ where the mass distribution actually exists. At this surface, the fields must satisfy the Maxwell equations with the mass source present, but we can also ask: what effective equations govern the field dynamics on the shell itself? By evaluating the field equations at the shell radius and integrating over the thin shell thickness, the delta function contributes a finite boundary term that modifies the effective equation. This yields the form presented below, where the mass term appears not as an external source, but as an intrinsic parameter in the dynamics of the fields on the shell surface. Therefore we drop the delta function and write:

$$\boldsymbol{\nabla} \times \mathbf{B}_S - \partial_t \mathbf{E}_S = -\frac{m}{\hbar}\mathbf{E}_S \tag{100}$$

Now, apply the duality transformation and write the solenoidal Maxwell equations in this form:

$$\begin{aligned} -\partial_t \mathbf{E}_S + \boldsymbol{\nabla} \times \mathbf{B}_S + \frac{m}{\hbar}\mathbf{E}_S = 0 \\ -\partial_t \mathbf{B}_S - \boldsymbol{\nabla} \times \mathbf{E}_S + \frac{m}{\hbar}\mathbf{B}_S = 0 \end{aligned} \tag{101}$$

Using the spinor curl definition:

$$\begin{aligned} -\partial_t \mathbf{E}_S + g_i \partial_i \mathbf{B}_S + \frac{m}{\hbar}\mathbf{E}_S = 0 \\ -\partial_t \mathbf{B}_S - g_i \partial_i \mathbf{E}_S + \frac{m}{\hbar}\mathbf{B}_S = 0 \end{aligned} \tag{102}$$

Now define a six-component field:

$$\psi = \begin{pmatrix} \mathbf{E}_S \\ \mathbf{B}_S \end{pmatrix} \tag{103}$$

We can combine the two equations above into a compact form:

$$\left(\gamma^\mu \partial_\mu + \frac{m}{\hbar}\right) \psi = 0 \tag{104}$$

This has the exact form of the Dirac equation, as given in Sakurai's Advanced Quantum Mechanics, Eq. 3.31.

The matrices g_i defined here are the standard generators of SO(3) rotations—the antisymmetric 3x3 matrices representing infinitesimal rotations in three-dimensional space. The representation $\boldsymbol{\nabla} \times \mathbf{V} = g_i \partial_i \mathbf{V}$ can be verified by direct computation: the curl operation naturally emerges from the commutation relations of these generators. While this is not the standard Dirac representation (which operates on four-component spinors in Minkowski space), it serves to express the Maxwell curl operations in a form that makes the structural similarity to the Dirac equation manifest. The resulting six-component field $\psi = (\mathbf{E}_S, \mathbf{B}_S)^T$ is not a quantum spinor but a classical field object—yet the mathematical parallelism is suggestive of deeper connections between classical electromagnetism and relativistic quantum mechanics that merit further investigation.

This is a remarkable result: starting from classical field equations and a single added hypothesis, we arrive at a spinor field satisfying the Dirac equation. This is of course done without quantization or any reference to spin.

9 Conclusion

Can gravity emerge from classical physics, without invoking curved spacetime or quantum gravity? That was the question we asked. Faced with two powerful but incompatible theories, quantum mechanics and general relativity, we chose a different path: to treat mass as a source of momentum flow, drawing from a pervasive background field. The result is a framework that is simple, classical, and deeply physical.

To test the idea, we built the theory step by step. We formulated equations for how mass interacts with the vacuum field. We introduced cutoffs to make the interaction finite and consistent. We checked for conservation of energy and momentum. Then we asked: does this theory match observation? And the answer, to leading order, is yes. It reproduces the familiar predictions of general relativity: the bending of light, gravitational redshift, and the precession of Mercury.

But something unexpected happened. When we re-expressed the field equations in a different form, they gave rise to the Dirac equation, the same equation that governs quantum particles like electrons. This wasn't assumed; it emerged. That a single classical field framework can describe gravity, electromagnetism, and quantum behavior is remarkable. It suggests these aren't separate layers of physics, but different faces of the same underlying structure. One set of equations that does it all.

This stands in contrast to general relativity, where mass is treated as a cold object sitting in empty space and curving spacetime around it

9.1 Where This Could Go

The assumption that mass sinks radiation is implicit in the Feynman Lectures. ... but we are making it our ”core” hypothesis?

Here, in the current iteration of this paper, we simply state the hypothesis and explore its consequences. In future drafts or subsequent work, we hope to derive this form directly from the two assumptions laid out in the next sections.

Several directions remain open:

- Expansion of the energy outside of the radiation zone As mentioned in the text, surrounding equation 56, it would be interesting to expand this energy equation to ouside of the radiation zone.

- Expansion of the precession of the perihelion Calculation If we went to higher order in this calculation, we could have a simple test to test the difference ... This was probably explained in the text.

- Removal of the Hypothesis As mentioned in the text **Section ?**, it is the belief of this author that we can actually derive the hypothesis from the assumptions. Then this theory wouldn't even need an hypothesis.

9.2 Final Thought

Physicists have long sought simplicity. Apparently the universe is built on simple laws. Einstein advised making a theory as simple as possible, but no simpler. When a single framework explains more with less, it earns our attention. Bringing electricity, magnetism, gravity, and the Dirac equation under one set of equations is a level of simplicity that deserves serious attention.

The history of physics is full of bold guesses. Some are wrong, some are premature, and some are foundational. This is just a guess. But it is a specific, testable, and mathematically structured one. If nothing else, it shifts the question. Gravity need not emerge from geometry. It might instead arise from the quiet hiss of radiation no one thought to listen to.

This is offered not as a conclusion, but as a beginning. Collaborators welcome.

Acknowledgments

This paper would not exist without the support of a few key people. First and foremost, I thank my wife, who has stood beside me for over two decades as I pursued this idea. Her patience, steadiness, and quiet encouragement through all the ups and downs have been unwavering.

I am also deeply grateful to my friend Michael Seidel, who, despite not having a background in physics, listened to me explain these ideas again and again over the years. His willingness to engage, ask questions, and simply be present helped me find clarity when I needed it most.

To both of them: thank you.

References